

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Pharm. Examination University Theory Examination November 2017

PH417 Pharmaceutical Analysis – III

Date: 27.11.2017, Monday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

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|------------|--|-----------|
| Q 1 | Attempt all questions. Each question is of one mark. | 20 |
| 1. | Which one of the following vibrations is the most energetic?
[A] Wagging
[B] Twisting
[C] Bending
[D] Stretching | |
| 2. | IR spectroscopy is generally used to determine
[A] Functional group
[B] Molecular mass
[C] % purity
[D] Types and number of Protons present | |
| 3. | Mercury lamp is used as radiation source in which technique?
[A] Calorimetry
[B] FTIR
[C] UV Spectroscopy
[D] Fluorimetry | |
| 4. | Which of the following is used to calibrate wavelength scale of UV/Visible Spectrophotometer?
[A] Polystyrene
[B] Holmium perchlorate
[C] Potassium dichromate
[D] Potassium chloride | |
| 5. | Which of the following solvent is commonly used in NMR?
[A] Chloroform
[B] Methanol
[C] Carbon tetrachloride
[D] Acetone | |
| 6. | Position of signal in NMR spectrum indicates
[A] Number of different kind of the proton present in different environment
[B] Electronic environment of each kind of proton
[C] Relative number of protons of each kind
[D] Number of neighbouring proton present | |
| 7. | Which of the following has the highest chemical shift value?
[A] CH ₃ I | |

- [B] CH_3Br
[C] CH_3F
[D] CH_3Cl
8. Spin quantum number (I) of ^{13}C is
[A] 0
[B] 1
[C] $1/2$
[D] $3/2$
9. M-18 peak in mass spectrum indicates loss of
[A] Hydroxyl group
[B] Amine group
[C] Methyl group
[D] Water molecule
10. TGA curve is a plot of
[A] Weight Vs Temperature
[B] Weight loss Vs Temperature
[C] Weight Vs Volume of Titrant
[D] Weight Vs Current
11. ICH guideline on analytical method validation is described under _____ section.
[A] Quality
[B] Safety
[C] Efficacy
[D] Multidisciplinary
12. Which of the following technique is an atomic spectroscopic technique?
[A] UV/Visible Spectroscopy
[B] IR Spectroscopy
[C] Spectrofluorimetry
[D] Flame Photometry
13. Which of the following electronic transition requires highest energy?
[A] n to π^*
[B] π to π^*
[C] n to σ^*
[D] σ to σ^*
14. The transition zone for Raman spectra is
[A] Between vibrational and rotational levels
[B] Between electronic levels
[C] Between magnetic levels of nuclei
[D] Between magnetic levels of unpaired electrons
15. In Raman Spectroscopy, Light _____ is studied.
[A] Absorbed
[B] Transmitted
[C] Scattered
[D] Diffracted
16. Removal of a single electron from a molecule results in the formation of
[A] Fragment ion
[B] Metastable ion
[C] Molecular ion
[D] Rearrangement ion
17. _____ lamp is used as radiation source in AAS.
[A] Tungsten
[B] Hollow cathode
[C] Xenon
[D] Helium
18. X-ray effect is based on
[A] Outer shell electron transition
[B] Inner shell electron transition
[C] Rotation of molecule
[D] None of the above

19. When molecule is excited by the absorption of radiation it soon returns to ground state by losing its excess of energy emitted as radiation. This phenomenon is known as :
 [A] Fluorescence
 [B] Hypsochromic shift
 [C] Electron Emission
 [D] Reflection of radiation
20. Interaction between radiation and matter is possible when
 [A] Both of them are of different energy
 [B] Both of them are of similar energy
 [C] Matter is at lower energy than radiation energy
 [D] None of above

SECTION – II

- Q 2** Attempt any **TWO** of the following
- A** State Beer's Law of UV/Visible Absorption Spectroscopy. What are the conditions of Beer's Law? Describe the factors for deviations from Beer's Law. 05
- B** Discuss in brief theory of NMR Spectroscopy. 05
- C** What is the basic principle of Mass Spectrometry? Explain. 05
- Q 3** Attempt any **TWO** of the following
- A** Define following terms: 05
1. Wavenumber
 2. Chromophore
 3. Auxochrome
 4. Bathochromic shift
 5. Fermi Resonance
- B** Comment on the following statements: 05
1. Absorbance is unitless parameter.
 2. Line spectrum is observed in Atomic Spectroscopy.
 3. Vacuum is required for the Mass spectrometer.
 4. Molecular ion peak is obtained at the highest m/z ration.
 5. TMS is used as reference standard in NMR.
- C** 1. State the probable frequency in IR Spectra, at which a peak is obtained for the following functional groups: 05
 (a) $\text{-C}\equiv\text{N}$, (b) -COOH , (c) -C=O , (d) -CHO , (e) -NH_2
 2. How will you differentiate following pairs using NMR spectroscopy.
 (a) Acetone and Ethyl acetate
 (b) Acetic acid and Methanol

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Write ideal requirements and types of UV/Visible detectors. Explain ANY ONE detector in detail. 05
- B** Discuss applications of UV/Visible Spectroscopy. 05
- C** Discuss the working principle of differential scanning calorimetry with a suitable diagram. 05
- D** Draw schematic diagram of the following with labeling 05
 (1) Atomic Fluorescence Spectrometer
 (2) Double Beam UV/Visible Spectrometer
- E** What is Chemical Shift? Discuss factors affecting chemical shift. 05
- F** Write in detail about applications of Atomic Spectroscopy. 05

- Q 5** Attempt any **FOUR** of the following
- A** Describe the sample handling techniques of IR Spectroscopy. **05**
- B** Draw and describe the principle of Chemical ionisation source in Mass Spectrometer. Justify: CI is known as Soft Ionisation source. **05**
- C** Write a note on factors affecting the fluorescence intensity of a substance. **05**
- D** Describe the principle and origin of Raman Spectra with suitable diagram. State the advantages of Raman Spectroscopy over IR Spectroscopy. **05**
- E** Explain the principle of X-Ray Diffraction techniques with suitable diagram. State in brief the major application of X-Ray Diffraction techniques for study of pharmaceuticals. **05**
- F** Why drugs are considered as special commodities? Differentiate QC and QA. **05**
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